

Marco Mo — AI Engineer | ML Engineer
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SUMMARY

AI/ML Engineer with experience building end-to-end ML systems for time-series and LLM-based applications. Skilled in designing scalable data and training pipelines, deploying models, and implementing RAG systems for real-world use cases.

PROJECTS

Fraud Detection ML Pipeline (IEEE-CIS)

- Developed an end-to-end ML pipeline on a 600k transaction dataset with 400+ features.
- Designed validation strategies to handle highly imbalanced data (3.5% fraud) and prevent data leakage.
- Built modular components for training, evaluation, experimentation, and FastAPI-based model serving.
- Ensured reproducible results via MLflow; automated workflows with Docker and GitHub Actions

Offline RAG System for Legal Contracts

- Developed a retrieval-augmented generation pipeline over 400+ legal documents.
- Implemented ingestion pipeline with hierarchical chunking and hybrid retrieval (vector search + BM25).
- Built LLM-based querying system with local deployment using Streamlit.
- Designed system for offline use with efficient document indexing and retrieval.

EXPERIENCE

Teaching Assistant — Data Challenges, Data Analytics, Calculus Eindhoven University of Technology (TU/e)

Sep 2021 – Jul 2025

- Monitored group progress and guided teams using SCRUM-style meetings.
- Diagnosed underlying issues from students' answers and coached toward solutions.
- Led question-driven guidance to improve problem-solving and project quality.

Technical Lead – Student Team SimEnergy Eindhoven University of Technology (TU/e)

Sep 2021 – Jul 2023

- Developed machine learning models for appliance-level energy disaggregation using time-series data.
- Built end-to-end pipelines for model training and forecasting across 6+ appliance types.
- Translated model outputs into dashboards for use in energy-focused events.

EDUCATION

MSc Data Science & Artificial Intelligence | Eindhoven University of Technology

2023 – 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration using Different Entropy Measures.

Honors Academy, AI Track | Eindhoven University of Technology

2021 – 2023

- Projects: Energy disaggregation, GAN image generation.

BSc Data Science | Eindhoven University of Technology & Tilburg University

2020 – 2023

- Thesis: Medical AI Platform Text Classification.

SKILLS

Programming: Python, SQL, C++, R, Java

ML & Data: pandas, NumPy, scikit-learn, PyTorch, TensorFlow

LLM / GenAI: LangChain, Ollama, ChromaDB, Qdrant

Data Engineering / MLOps: Spark, MLflow, Docker, CI/CD (GitHub Actions), FastAPI

Visualization: Power BI, Plotly, Dash

Tools: Git

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)